

GREEN CHEMISTRY

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INTRODUCTION:

The design of processes that reduce or eliminate the use and production of toxic products is known as green chemistry. Green chemistry is also known as sustainable chemistry.

The term was first used by Paul T. Anastas [father of Green Chemistry] in the last decade of the 20th century

Green Chemistry implies:

- Prevention of pollution rather than treatment of pollution
- Environmentally Benign Chemistry
- Sustainable Chemistry
- Ecofriendly Chemistry
- Clean Chemistry

Green chemistry should not be confused with environmental chemistry as environmental chemistry deals with various facets of pollution, degree of pollution and treatment of pollution, while as green chemistry does not lead to pollution at all, hence we say it prevents pollution.

Thus green chemistry approach is a prevention approach, while as environmental chemistry approach is a treatment approach. Since prevention is better than cure, we may say green chemistry is better than environmental chemistry

Green chemistry is generally aimed at:

- Producing chemicals which are safe for biotic as well as abiotic environment.
- Using cost and energy effective methods and procedures.
- Designing processes that reduce or eliminate the use and production of toxic materials
- Minimizing the production of wastes.
- Avoiding the production of non-biodegradable materials/products.
- Maximizing the use of raw-materials from renewable resources

12 PRINCIPLES OF GREEN CHEMISTRY:

1. Prevention
2. Atom Economy
3. To avoid the use and formation to toxic materials
4. Use of nontoxic chemical products
5. Minimum use of auxiliary substance
6. Minimum energy consumption
7. Use of renewable sources
8. Minimization of steps
9. Use of catalytic reagents
10. Life time of a chemical product
11. Monitoring the generation of hazardous substance
12. Use of chemical safer substances

1. PREVENTION (HEART OF GREEN CHEMISTRY):

- It is better to prevent the formation of waste than to treat or clean up the waste after the formation.
- Use waste products of one system as the raw material for other systems.
- In the chemical synthesis, waste products should be zero or minimum.
- The first principle aims to develop the zero waste technology.
- if discharged causes pollution and requires expenditure for cleaning-up
- **EXAMPLE:** Bottom ash of thermal power stations can be used as a raw material for cement and brick industry.

2. ATOM ECONOMY:

- During the synthesis of a chemical product, the methodology should be designed in the way to maximize the incorporation of starting materials into the desired final product
- Thus it demands to minimize the formation of by-product
- **EXAMPLE:** The **Haber-Bosch process** ($\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$) has high atom economy because nearly all reactant atoms end up in the product (ammonia). In the chemical synthesis, waste products should be zero or minimum.

3. TO AVOID THE USE AND FORMATION TO TOXIC MATERIALS:

- If possible and the synthetic methodologies should avoid the use and generation of toxic and environmentally hazardous substances.
- This principle aims to develop the methodologies that will minimize the use and formation of toxic and hazardous substances.
- **EXAMPLE:** Replacing **phthalate plasticizers** (toxic) with **citrate-based plasticizers** (non-toxic) in PVC products.

4. USE OF NONTOXIC CHEMICAL PRODUCTS:

- Chemical products aims to be used in different activities should have the efficacy to function but with reduced environment.
- In many chemical industries, not only the waste product but the starting materials are also quite hazardous to the workers and environment.
- **EXAMPLE:** Using **supercritical CO₂** (non-toxic) instead of **organic solvents** (toxic) for dry cleaning.

5. MINIMUM USE OF AUXILIARY SUBSTANCES:

- In a chemical synthesis ,the use of auxiliary substance like solvents, separating agents,etc,should be avoided. If, these are to be used, they should be ecofriendly.
- This principle aims to use of green solvents(e.g, water, supercritical CO₂)in the place of volatile halogenated organic solvents eg. CH₃Cl₃,CHCl₃,C₂Cl₄,CCl₄ for chemical synthesis and other purposes.
- **EXAMPLE:** Claisen rearrangement.

6. MINIMUM ENERGY CONSUMPTION:

- In the synthesis of a chemical product, the energy, the energy consumption should be minimized to make the process more and more economic, Ideally, the synthetic methods be carried out at ambient temperature and pressure.
- To save energy, synthetic methodologies should needs more and more moderate conditions and the ambient temperature and pressure are the best choices.
- It needs suitable catalysts that will accelerate the reaction rate even at lower temperature. The biocatalysts can work at the ambient conditions.
- **EXAMPLE:** Using **microwave-assisted synthesis** to reduce reaction time and energy compared to conventional heating.

7. USE OF RENEWABLE SOURCES:

- If it is technically and economically possible, then the renewable resources(e.g. biomass) rather than the non renewable recourses (e.g. crude oil)should be used as the raw material or feed stock.
- Continuous use of the nonrenewable feedstock (e.g. petroleum product, fossil fuel) will deplete the resource and future generation will be deprived. Moreover, use of the these nonrenewable resources puts a burden on the environment.
- **EXAMPLE:** Producing **polylactic acid (PLA) plastic** from corn starch instead of petroleum-based plastics.

8. MINIMIZATION OF STEPS:

- The steps blocking groups, production/reproduction of the group, temporary modification of physical and chemical processes etc., should be avoided as far as possible during the synthesis of a chemical product. Thus there should be a minimum of steps to synthesize a target product.
- Specially in organic synthesis need very often protection of some functional groups.
- **EXAMPLE:** A **one-pot synthesis** where multiple reactions occur in a single vessel, reducing waste and time.

9. USE OF CATALYTIC REAGENTS:

- Selective catalytic reagents are superior to stoichiometric reagents in a chemical synthesis. This will save the energy and reduce the burden of by-product.
- This principle of green chemistry states that catalytic of the following advantages.
 - (i) 100% atom economy because the true catalysts are fully recovered without any change in their chemical and physical properties.
 - (ii) the catalyzed reactions are faster i.e. energy save possible.
 - (iii) reaction yields are better.
 - (iv) Selective reaction products.
 - (v) maximum utilization of the starting material and minimum production of the waste material.
- **EXAMPLE:** Using **enzymes (biocatalysts)** in biodiesel production instead of strong acids/bases.

10. LIFETIME OF A CHEMICAL PRODUCT (DESIGN DEGRADATION):

- At the end of function, the chemical products (e.g. pesticides) should degrade easily to harmless products, i.e. after their function, they should not persist in the environment.
- DDT is the best example. It is an effective pesticide but its stability in the natural environment causes several environment hazards.
- It states that the waste product should degrade automatically to clean the environment.
- **EXAMPLE:** Designing **biodegradable polymers** (e.g., PHA) that break down naturally instead of persisting like traditional plastics.

11. MONITORING THE GENERATION OF HAZARDOUS SUBSTANCES:

- Analytical methodologies should be further developed to allow for real time in process monitoring and control prior to the generation of hazardous substance in the synthesis of chemical products.
- analytical methodologies should be developed or modified, so that continuous monitoring of the manufacturing and processing units is possible.
- **EXAMPLE:** Using **real-time sensors** in chemical plants to detect and minimize hazardous byproducts like dioxins.

12. USE OF CHEMICAL SAFER SUBSTANCES:

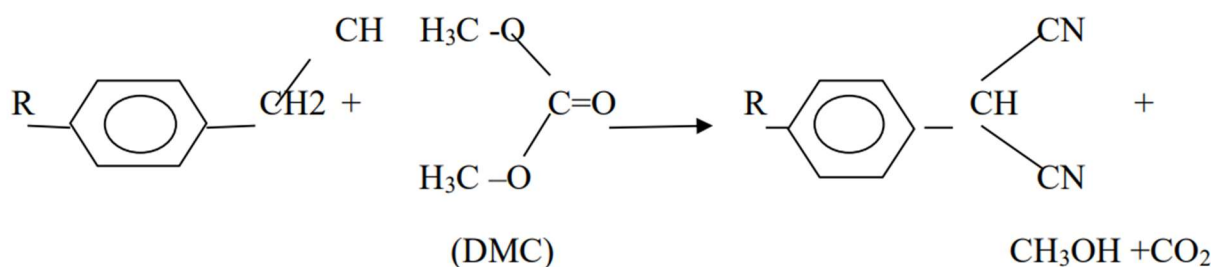
- The substances to be used in a chemical reaction should be selected in a way that they can minimize the occurrence of chemical accidents, explosions, fires and emissions. In other words, the substance to be used should not be hazardous.
- : The substance used in chemical industries should be in such forms so that the possibility of accidents can be minimized.
- **EXAMPLE:** Replacing **volatile organic solvents (VOCs)** with **water-based solvents** in paints.

SOME IMPORTANT TECHNIQUES AND DIRECTION IN PRACTISING GREEN CHEMISTRY:

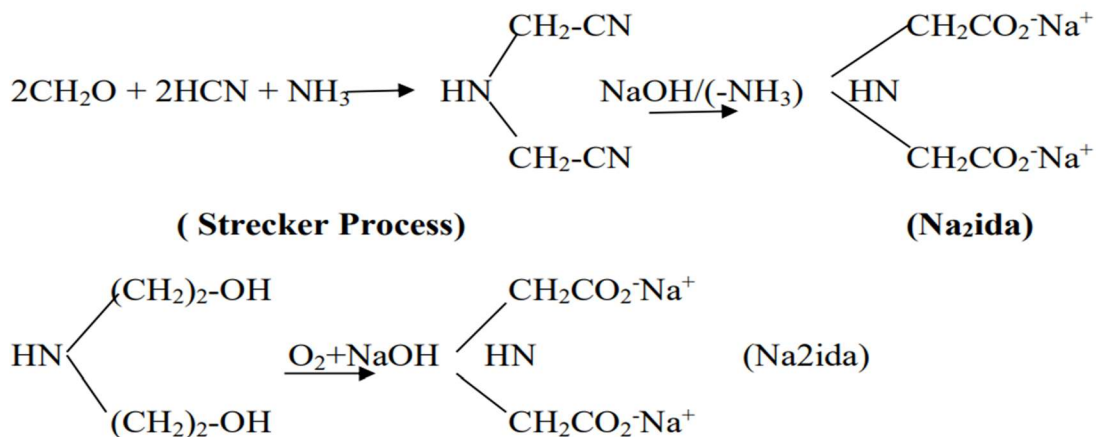
1. USE OF SAFER REAGENTS:

Green chemistry aims to replace the hazardous reagents. It is illustrated for the methylation reaction which conventionally uses the hazardous reagents like $\text{CH}_3\text{-X}$ (methyl halide), DMS (dimethylsulfate), etc. The green alternative is DMC (dimethylcarbonate).

Methylation of an active methylene group can be done by using the green reagents

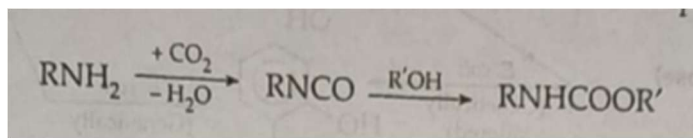


The Conventional methylation process produces the hazardous toxic by product while in the above methylation reaction, the reagents are green and the by products are also green. Disodium iminodiacetate is an essential reagent in manufacturing a herbicide. Conventional route uses the hazardous chemicals like formaldehyde and HCN while the green synthesis does not use any such hazardous substance.



(Diethanol amine) (Green Synthesis)

K₂CO₃ is also a green reagent that finds uses in the synthesis of different heterocyclic compounds. It acts as a base and provides a solvent free condition in many cases. Conventional synthesis of polyurethanes needs the use of the toxic chemical like phosgen gas. The green technique avoids the use of phosgen gas.



2. USE OF GREEN SOLVENT:

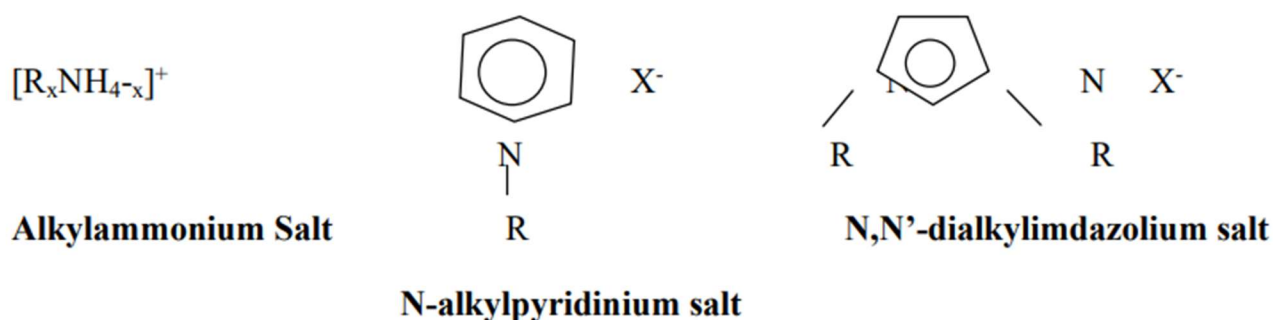
- Green solvents are **renewable, biodegradable, and less toxic**
- While non-green solvents are often **petroleum-based, hazardous, or persistent pollutants**.
- In Microwave assisted organic reactions green solvents are playing important roles in green chemistry.
- Organic synthesis in aqueous media is becoming increasingly popular. Different organic reactions in aqueous media coupled with microwave/ultrasonic irradiation have now been established.
- Supercritical water in which organic compounds are more soluble compound to water at ordinary condition is also a potential alternative.
- **Green Solvents-**
 - Water
 - Supercritical CO₂ (scCO₂)
 - Ionic Liquids
 - Ethanol (Bio-based)
 - Glycerol
 - 2-Methyltetrahydrofuran (2-MeTHF)
 - Cyrene (Dihydrolevoglucosenone)
 - Deep Eutectic Solvents (DES)
 - Limonene (from citrus peels)
 - Polyethylene Glycol (PEG)

- **Non-Green Solvents-**

Benzene (carcinogenic)
 Chloroform (toxic, ozone-depleting)
 Carbon Tetrachloride (highly toxic)
 Diethyl Ether (highly flammable)
 Dimethylformamide (DMF, toxic)
 N-Methyl-2-pyrrolidone (NMP, reproductive toxin)
 Tetrahydrofuran (THF, petroleum-based, flammable)
 Hexane (neurotoxic, petroleum-based)
 Toluene (volatile, toxic)
 Xylene (hazardous, petroleum-based)

3. USE OF ROOM TEMPERATURE IONIC LIQUIDS AS GREEN SOLVENTS:

Commonly used organic solvents are generally volatile organic compounds (VOCs) that cause several environmental hazards including smog production in air, ozone depletion, health hazards etc. The ionic liquids are the green alternatives. These ionic liquids are generally organic salts like alkylammonium salts, N-alkylpyridinium salts, and N,N'-dialkylimidazolium salts.



These nonvolatile ionic liquids can be easily recovered after use. They can be recovered and purified. The product can be isolated from the ionic liquid by using some common solvent like either in a separating funnel.

4. SUPERCRITICAL FLUIDS AS GREEN SOLVENTS:

Use of supercritical CO₂ fluids as green solvents: Use of supercritical CO₂ emerging with a great substitute the VOC's. The important uses of SC-CO₂ are: dry cleaning of clothes, polymerisation reaction etc. Supercritical CO₂ may be used to extract the organic pollutants like MTBE, PCBs, DDT etc. present in water, soils and sediments. Supercritical water (T_c=374°C) is too hot for many organic compounds. But near critical water appears as a benign solvent for many organic reactions.

5. USE OF GREENER CATALYST:

Some times the used catalysis are toxic and these are accumulated in the industrial waste product. For example HF is the catalyst used in the production of linear alkylbenzenes and it has been replaced by the greener catalyst i.e., fluorided silica alumina catalyst. Other example of greener catalyst are zeolites acting as acid catalysts in Friedle crafts reaction, synthesis of oxime by titanium(IV) –silicate (TS-1) catalyst, dehydration of organic molecules etc by ZSN-5 (used in the manufacture of synthetic gasoline, TiO₂ (used as a photocatalyst in removing water pollutants. Biocatalysts are the most ideal, green catalysts and these are finding applications in many biocatalytic transformations. These biocatalysts may be developed in the genetically engineered bacteria.

6. ORGANIC WASTE MANAGEMENT:

Oxidation of organic compounds is often required in both organic synthesis and the treatment of waste from chemical industry effluents. Traditionally, metal complex catalysts are used for these reactions, which can impose a significant environmental burden. Green technology addresses this issue by employing the **laccase enzyme**, which oxidizes many organic compounds in aqueous media using only atmospheric oxygen under ambient conditions.

1 MARK MCQ:

1. Who is known as the 'Father of Green Chemistry'?

- a) Marie Curie
- b) Paul T. Anastas
- c) Robert Boyle
- d) Linus Pauling

Answer: b) Paul T. Anastas

2. Which principle is considered the 'Heart of Green Chemistry'?

- a) Atom Economy
- b) Prevention
- c) Use of Catalysts
- d) Renewable Resources

Answer: b) Prevention

3. The Haber-Bosch process ($N_2 + 3H_2 \rightarrow 2NH_3$) is an example of:

- a) Low atom economy
- b) High atom economy
- c) Toxic byproduct formation
- d) Non-renewable energy use

Answer: b) High atom economy

4. Which of the following is a green alternative to phthalate plasticizers in PVC?

- a) Benzene
- b) Citrate-based plasticizers
- c) DDT
- d) Formaldehyde

Answer: b) Citrate-based plasticizers

5. Supercritical CO₂ is used as a green solvent in:

- a) Dry cleaning
- b) Pesticide synthesis
- c) Lead-based paints
- d) Nuclear waste treatment

Answer: a) Dry cleaning

6. Which solvent is NOT green?

- a) Water
- b) Ethanol (bio-based)
- c) Benzene
- d) Ionic liquids

Answer: c) Benzene

7. Polylactic acid (PLA) is derived from:

- a) Petroleum
- b) Corn starch
- c) Coal tar
- d) Natural gas

Answer: b) Corn starch

8. Microwave-assisted synthesis is preferred because it:

- a) Increases energy consumption
- b) Reduces reaction time and energy
- c) Requires toxic solvents
- d) Generates more waste

Answer: b) Reduces reaction time and energy

9. The laccase enzyme oxidizes organic compounds using:

- a) Chlorine gas
- b) Atmospheric oxygen
- c) Heavy metals
- d) Sulfuric acid

Answer: b) Atmospheric oxygen

10. Which principle emphasizes designing chemicals to degrade after use?

- a) Atom Economy
- b) Design for Degradation
- c) Use of Catalysts
- d) Renewable Feedstocks

Answer: b) Design for Degradation

11. DMC (Dimethylcarbonate) is a green alternative for:

- a) Methylation reactions
- b) Polymerization
- c) Ozone depletion
- d) Waste incineration

Answer: a) Methylation reactions

12. Non-green solvents include:

- a) Supercritical CO₂
- b) Limonene
- c) Carbon tetrachloride
- d) Polyethylene glycol

Answer: c) Carbon tetrachloride

13. Zeolites are used as greener catalysts in:

- a) Friedel-Crafts reactions
- b) PVC synthesis
- c) DDT production
- d) Nuclear fusion

Answer: a) Friedel-Crafts reactions

14. The Strecker process uses which hazardous chemicals?

- a) Formaldehyde and HCN
- b) CO₂ and water
- c) Ethanol and glycerol
- d) Limonene and PEG

Answer: a) Formaldehyde and HCN

15. Ionic liquids are preferred over VOCs because they are:

- a) Highly volatile
- b) Non-volatile and recyclable
- c) Petroleum-based
- d) Radioactive

Answer: b) Non-volatile and recyclable

16. Which is a biodegradable polymer?

- a) Polyethylene
- b) PHA (Polyhydroxyalkanoates)
- c) PVC
- d) Polystyrene

Answer: b) PHA

17. Real-time monitoring in industries helps reduce:

- a) Product yield
- b) Hazardous byproducts
- c) Renewable resource use
- d) Catalyst efficiency

Answer: b) Hazardous byproducts

18. Water-based paints replace:

- a) VOCs (Volatile Organic Compounds)
- b) Ionic liquids
- c) Supercritical fluids
- d) Zeolites

Answer: a) VOCs

19. The primary goal of green chemistry is to:

- a) Increase waste production
- b) Eliminate toxicity at the design stage
- c) Use more fossil fuels
- d) Avoid catalysis

Answer: b) Eliminate toxicity at the design stage

20. K_2CO_3 is a green reagent because it:

- a) Is highly toxic
- b) Provides solvent-free conditions
- c) Generates phosgene gas
- d) Is non-biodegradable

Answer: b) Provides solvent-free conditions

21. Which of the following is a key feature of green chemistry?

- a) Maximizing waste production
- b) Using non-renewable resources
- c) Preventing pollution at the source
- d) Increasing energy consumption

Answer: c) Preventing pollution at the source

22. The replacement of phosgene gas in polyurethane synthesis is an example of:

- a) Atom economy
- b) Use of safer reagents
- c) Non-green chemistry
- d) Increased toxicity

Answer: b) Use of safer reagents

23. Deep Eutectic Solvents (DES) are considered green because they are:

- a) Petroleum-derived
- b) Toxic and volatile
- c) Biodegradable and non-toxic
- d) Used only in high-energy processes

Answer: c) Biodegradable and non-toxic

24. Which principle advocates for fewer synthetic steps in a process?

- a) Minimization of steps
- b) Use of renewable sources
- c) Lifetime of a chemical product
- d) Monitoring hazardous substances

Answer: a) Minimization of steps

25. The term "supercritical fluid" refers to a substance that is:

- a) Below its critical temperature and pressure
- b) At its critical temperature and pressure
- c) Only a solid state
- d) Exclusively a gas

Answer: b) At its critical temperature and pressure

26. Which of the following is a renewable feedstock?

- a) Crude oil
- b) Natural gas
- c) Biomass
- d) Coal

Answer: c) Biomass

27. The main advantage of catalytic reagents over stoichiometric reagents is:

- a) Higher waste generation
- b) Lower atom economy
- c) Reusability and reduced waste
- d) Increased energy consumption

Answer: c) Reusability and reduced waste

28. Titanium(IV)-silicate (TS-1) is a greener catalyst used in:

- a) Oxime synthesis
- b) PVC production
- c) DDT synthesis
- d) Nuclear reactions

Answer: a) Oxime synthesis

29. Which of the following is a volatile organic compound (VOC)?

- a) Water
- b) Supercritical CO₂
- c) Toluene
- d) Ionic liquids

Answer: c) Toluene

30. The primary environmental issue with DDT is its:

- a) Rapid degradation
- b) Persistence in the environment
- c) Low toxicity
- d) Renewable source

Answer: b) Persistence in the environment

31. Microwave-assisted reactions align with green chemistry by:

- a) Increasing solvent use
- b) Reducing energy and time
- c) Generating more byproducts
- d) Using toxic catalysts

Answer: b) Reducing energy and time

32. Which compound is a byproduct of biodiesel production and used as a green solvent?

- a) Benzene
- b) Glycerol
- c) Phosgene
- d) Carbon tetrachloride

Answer: b) Glycerol

33. The function of laccase enzyme in waste treatment is to:

- a) Synthesize plastics
- b) Oxidize organic compounds using O₂
- c) Generate toxic metals
- d) Increase fossil fuel use

Answer: b) Oxidize organic compounds using O₂

34. Which of the following is NOT a green solvent?

- a) 2-Methyltetrahydrofuran (2-MeTHF)
- b) Dimethylformamide (DMF)
- c) Cyrene
- d) Ethanol (bio-based)

Answer: b) Dimethylformamide (DMF)

35. The green chemistry alternative to the Strecker process avoids:

- a) Use of formaldehyde and HCN
- b) Use of CO₂
- c) Use of renewable energy
- d) Use of biodegradable solvents

Answer: a) Use of formaldehyde and HCN

36. Zeolites are used as catalysts because they:

- a) Are toxic and non-recyclable
- b) Have high selectivity and reusability
- c) Increase waste production
- d) Require high-energy conditions

Answer: b) Have high selectivity and reusability

37. The term "atom economy" refers to:

- a) Maximizing byproduct formation
- b) Incorporating most reactants into the final product
- c) Using excess reagents
- d) Ignoring waste generation

Answer: b) Incorporating most reactants into the final product

38. Near-critical water is used as a solvent because it:

- a) Is highly toxic
- b) Dissolves organic compounds better than regular water
- c) Requires fossil fuels
- d) Generates persistent pollutants

Answer: b) Dissolves organic compounds better than regular water



39. Which of the following is a biocatalyst?

- a) HF (Hydrofluoric acid)
- b) Zeolite
- c) Laccase enzyme
- d) Titanium dioxide

Answer: c) Laccase enzyme

40. The primary goal of using real-time sensors in industries is to:

- a) Increase hazardous byproducts
- b) Monitor and minimize hazardous substances
- c) Reduce product yield
- d) Avoid using catalysts

Answer: b) Monitor and minimize hazardous substances